

Zhihong Zhang

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Data Engineer × AI Engineer — Industrial IoT · Predictive Maintenance · Supply Chain Analytics · GenAI Agents

Summary

Data & AI engineer with Chemical Engineering and Mathematics background. Built scalable IoT data pipelines and predictive-maintenance platforms for hydrogen refueling stations; anomaly response time down 80%, equipment failure rate down 35%. Strong in Python/SQL/TensorFlow, real-time dashboards, and AI-assisted automation. Bilingual Mandarin/English; eight years in the US.

Experience

Sinopec Hydrogen Energy (Shanghai) Shanghai, China
IT Operations & Data Analyst Jan 2023 – Jun 2026

- Built real-time IoT data pipelines that collect multi-source sensor data from hydrogen refueling stations; built data platforms and operational dashboards that turn time-series metrics into actionable maintenance decisions.
- Developed real-time equipment-health dashboards for KPI monitoring, reducing equipment failure rate by ~35%.
- Led cross-functional collaboration with university researchers to develop predictive-maintenance models for diaphragm compressors under scarce-failure, low-SNR conditions; built TensorFlow models fused with equipment failure-mode knowledge for fault prediction.
- Analyzed daily/weekly station operations data to optimize compressor scheduling and pressurization strategies, balancing load and energy consumption while reducing per-kg hydrogen energy cost by ~12%.
- Adopted AI-assisted development tools (Claude Code, multi-agent workflows) to automate data-analysis and reporting workflows, improving routine development efficiency by ~3× and reducing manual repetitive work.

Freelance / Independent Consulting Shanghai, China
Data Analysis & Financial Modeling Feb 2021 – Dec 2022

- Built automated data pipelines and dashboards for 2–3 early-stage startups and individual clients, enabling KPI tracking, portfolio risk assessment, and data-driven decision workflows.
- Provided market time-series signal research and backtesting framework development, applying anomaly-detection and noise-filtering methods from industrial signal processing to financial data.
- Helped clients define core business metrics, build dashboards, and establish repeatable data-driven reporting for business decisions.

Yeshiva University New York, USA
Data Science Graduate Project Aug 2019 – Jan 2021

- Capstone: optimized a PyTorch CNN for audio time-series speaker verification, completing end-to-end feature extraction and denoising, reducing EER by 10%.
- Insurance risk prediction on imbalanced datasets using scikit-learn and TensorFlow, improving accuracy from 73% to 88%.

Selected Projects

- **H2 Station Anomaly Early-Warning System [response time ↓80%]** — multi-sensor real-time pipeline, anomaly detection & tiered alerting (*Proprietary; architecture and alerting-strategy walkthrough available in interview*)
- **Compressor Predictive Maintenance [failure rate ↓35%]** — fault prediction under scarce-sample, low-SNR conditions: mechanism-informed features × TensorFlow
- **Market Time-Series Signal Research (personal) [in progress]** — transferring industrial anomaly-detection methods to financial time-series: noise filtering and signal significance

Skills

Data Engineering & Analytics	Python, SQL, Industrial Data Pipelines, Data Cleaning, Feature Engineering, Time-Series Analytics, KPI Monitoring, Dashboard Development, Tableau
AI & Machine Learning	TensorFlow, scikit-learn, Predictive Maintenance, Anomaly Detection, Machine Learning, AI-assisted Development, Multi-Agent Workflow Automation
Cloud & Infrastructure	AWS (EC2, RDS, Lambda), Docker, Linux, Jenkins CI/CD, edge-to-cloud streaming
Industrial Domain	Industrial IoT, Manufacturing Analytics, Equipment Health Monitoring, Hydrogen Refueling Stations, Failure Mode Analysis, Statistical Modeling, Process Optimization

Education

Yeshiva University	2019 – 2021, New York, USA
M.S. Data Analytics and Visualization (GPA 3.9/4.0)	
Calvin University	2012 – 2017, Michigan, USA
B.S. Mathematics (Minors: Engineering, Chemistry)	
Technical University of Berlin (Summer School)	Summer 2013, Berlin, Germany
Civil Engineering Statics & Dynamics, German	